

Adding a Slow Recovery Variable to the Yuval Neuron Model

C. elegans Locomotion Circuit

Distribution of Oscillators Project

April 14, 2026

Abstract

The piecewise-linear Yuval neuron model is intrinsically bistable: with zero applied current, the plateau $V = H$ is a stable fixed point and the neuron cannot return to rest without an external input. This document motivates and derives a minimal extension — a slow intrinsic recovery variable w — that converts the bistable system into a limit-cycle oscillator, eliminating plateau locking while preserving the qualitative dynamics required for the Distribution of Oscillators framework.

1 The Problem: Structural Bistability

Each neuron in the current model satisfies

$$C \dot{V} = g f(V) - I_{\text{syn}} + I_{\text{app}}, \quad (1)$$

where $f(V)$ is the piecewise-linear membrane nonlinearity:

$$f(V) = \begin{cases} m_1(-V + L) & V < L \\ m_2(V - L)(V - T) & L \leq V \leq T \\ m_3(V - H)(V - T) & T < V \leq H \\ m_4(-V + H) & V > H. \end{cases} \quad (2)$$

By construction, $f(L) = 0$ and $f(H) = 0$ for any choice of parameters. When $I_{\text{app}} = 0$ and $I_{\text{syn}} = 0$, the fixed points of (1) are exactly the zeros of $f(V)$: rest ($V = L$), threshold ($V = T$, unstable), and plateau ($V = H$). Both L and H are stable, so the neuron is *bistable by construction*.

Synaptic fatigue (the SF variable) addresses this in a *network* context: a fatiguing inhibitory partner can pull the active neuron off the plateau. However, an isolated neuron receiving $I_{\text{app}} = 0$ has no such partner and will remain locked at the plateau indefinitely.

Attempts to resolve this by pushing T close to H shrink the basin of attraction of the plateau but do not eliminate it, because $f(H) = 0$ is a structural property of the piecewise-linear form, not a consequence of any particular parameter choice.

2 The Fix: A Slow Recovery Variable

The standard approach in computational neuroscience is to augment the voltage equation with a slow variable w that plays the role of an intrinsic recovery current. The extended system is

$$C \dot{V} = g f(V) - w - I_{\text{syn}} + I_{\text{app}}, \quad (3)$$

$$\tau_w \dot{w} = w_{\infty}(V) - w, \quad (4)$$

where $\tau_w \gg \tau_V$ (slow timescale) and $w_{\infty}(V)$ is a monotonically increasing target function. Three natural choices for $w_{\infty}(V)$ are discussed in Section 3.

2.1 Geometric Interpretation

The V -nullcline ($\dot{V} = 0$) is now

$$w = g f(V) + I_{\text{app}}, \quad (5)$$

which is an N -shaped curve inherited from $f(V)$. The w -nullcline ($\dot{w} = 0$) is

$$w = w_{\infty}(V), \quad (6)$$

a monotonically increasing curve. The system has a fixed point wherever the two nullclines intersect.

When $I_{\text{app}} = 0$, the V -nullcline passes through $(L, 0)$ and $(H, 0)$. If w_{∞} is chosen so that its curve intersects the V -nullcline *only on the left (rest) branch*, then the plateau fixed point is destroyed and the only attractor is rest. For larger I_{app} , the V -nullcline shifts upward, the intersection can move to the right branch, and the system either settles at a depolarized fixed point or — if the intersection is in the negative-slope region of $f(V)$ — executes a stable limit cycle (oscillation).

This is exactly the FitzHugh–Nagumo geometry applied to the piecewise-linear model.

3 Choices for $w_{\infty}(V)$

3.1 Linear (simplest)

$$w_{\infty}(V) = \alpha (V - L), \quad (7)$$

with $\alpha > 0$. At rest $V = L$, so $w_{\infty} = 0$ and $w \rightarrow 0$: the recovery variable vanishes at rest. At the plateau $V = H$, $w_{\infty} = \alpha(H - L) > 0$, providing a net hyperpolarizing current that pulls the neuron off the plateau. This is the most transparent choice and introduces a single free parameter α .

The condition for the plateau fixed point to be eliminated is that the w -nullcline evaluated at H lies *above* the V -nullcline at H (which equals zero):

$$\alpha(H - L) > 0, \quad (8)$$

which is always satisfied for $\alpha > 0$. Any positive α eliminates the plateau fixed point when $I_{\text{app}} = 0$.

3.2 Sigmoid (biophysically motivated)

$$w_{\infty}(V) = \frac{w_{\max}}{1 + e^{-k_w(V - V_w)}} \quad (9)$$

A sigmoid centered near T or above mimics slow K^+ channel activation. This is the biophysically natural form but introduces three parameters ($w_{\max}$, k_w , V_w), most of which are not constrained by existing data for this model.

3.3 Threshold-gated (piecewise, consistent with existing style)

$$w_{\infty}(V) = \begin{cases} 0 & V \leq T \\ \beta(V - T) & V > T, \end{cases} \quad (10)$$

Recovery is zero below threshold and grows linearly above it. This is consistent with the piecewise-linear style of the rest of the model and keeps the number of free parameters minimal (one: β). At rest $w_{\infty}(L) = 0$; at the plateau $w_{\infty}(H) = \beta(H - T) > 0$.

4 Timescale Separation and Oscillation

For the extended system to produce oscillations, the recovery variable must be slow enough that the voltage can traverse the active phase before w equilibrates, but fast enough that w accumulates meaningfully during a plateau. Concretely:

- $\tau_w \ll$ plateau duration: w builds up during the plateau and terminates it.
- $\tau_w \gg$ spike upswing: the upswing is fast enough that w does not prevent activation.

A rough estimate: plateau duration is set by the synaptic fatigue timescale ($1/a \approx 1/0.000035 \approx 28,000$ ms per unit drive), while the membrane time constant is $C/g = 7$ ms. A reasonable starting range for τ_w is 100–1000 ms, which is slow relative to the membrane but fast relative to the network rhythm.

5 Interaction with Synaptic Fatigue

The existing synaptic fatigue variable SF acts as a slow recovery variable at the *network* level: it terminates inhibition from the active neuron, allowing its partner to escape. The new w variable acts at the *cell* level: it terminates the plateau of the active neuron directly.

These two mechanisms are complementary:

Mechanism	Acts on	Timescale
Synaptic fatigue SF	Inhibitory current onto partner	$1/a, 1/b$ (very slow)
Recovery variable w	Active neuron’s own plateau	τ_w (tunable)

With both mechanisms present, the half-center oscillator has two routes to burst termination. This makes the oscillation more robust and reduces dependence on the partner neuron’s fatigue — a single isolated neuron can now oscillate intrinsically when $I_{\text{app}} > 0$, and return to rest when $I_{\text{app}} = 0$.

6 Implementation in the Existing Codebase

The minimal change to `src/functions.py` is to add the w -dynamics:

$$\dot{w} = \frac{w_{\infty}(V) - w}{\tau_w}, \quad (11)$$

and to modify `src/network.py` to expand the state vector from $[V_1, \dots, V_N, \text{SF}_1, \dots, \text{SF}_N]$ to $[V_1, \dots, V_N, \text{SF}_1, \dots, \text{SF}_N, w_1, \dots, w_N]$, subtracting w_i from the $\dot{V}_i$ equation.

New global parameters to add to `src/settings.py`:

- `tau_w`: recovery timescale (ms), suggested initial value 200 ms.
- `w_scale`: amplitude scaling (α or β above), suggested initial value 0.01 nS/mV.
- `V_w`: threshold for the piecewise variant ($= T$ in the simplest case).

7 Summary

The bistability of the Yuval model is structural: $f(H) = 0$ by construction, making the plateau a permanent fixed point when $I_{\text{app}} = 0$. A slow recovery variable w resolves this by adding a nullcline that intersects $f(V)$ only on the rest branch at zero drive. The piecewise-linear form $w_{\infty}(V) = \beta \max(V - T, 0)$ is the most natural extension consistent with the existing model style, introduces one free parameter, and ensures the neuron returns to rest under zero drive while remaining capable of oscillation under sufficient excitation.